FUEL RAIL/FUEL DAMPER - 2.4L

DESCRIPTION - 2.4L

The fuel injector rail is used to mount the fuel injectors to the engine (Fig. 4). On the 2.4L 4-cylinder engine, a **fuel damper** is located near the front of the fuel rail (Fig. 4).

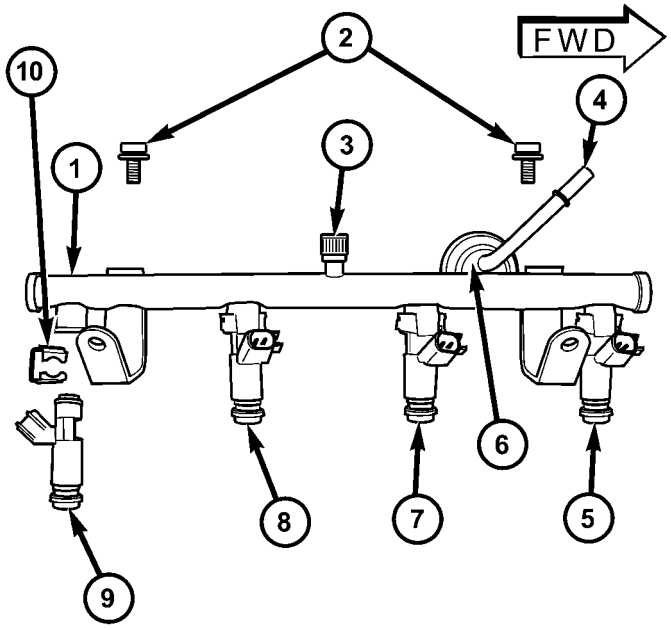

Fig. 4 FUEL RAIL - 2.4L

1 - FUEL RAIL
2 - MOUNTING BOLTS
3 - TEST PORT (IF USED)
4 - QUICK-CONNECT FITTING
5 - INJ. #1
6 - DAMPER
7 - INJ #2
8 - INJ #3
9 - INJ #4
10- INJECTOR RETAINING CLIP

OPERATION - 2.4L

The fuel injector rail supplies the necessary fuel to each individual fuel injector.

The fuel damper is used only to help control fuel pressure pulsations. These pulsations are the result of the firing of the fuel injectors. It is **not used** as a fuel pressure regulator. The fuel pressure regulator is **not mounted** to the fuel rail on any engine. It is located on the fuel tank mounted fuel pump module. Refer to Fuel Pressure Regulator for additional information.

The fuel rail is not repairable.

A quick-connect fitting with a safety latch is used to attach the fuel line to the fuel rail.

REMOVAL - 2.4L

WARNING: THE FUEL SYSTEM IS UNDER CONSTANT PRESSURE EVEN WITH ENGINE OFF. BEFORE SERVICING FUEL RAIL, FUEL SYSTEM PRESSURE MUST BE RELEASED.

The fuel rail can be removed without removing the intake manifold if the following procedures are followed.

(1) Remove fuel tank filler tube cap.
(2) Perform Fuel System Pressure Release Procedure.
(3) Remove negative battery cable at battery.
(4) Remove air duct at throttle body.
(5) Disconnect fuel line latch clip and fuel line at fuel rail. A special tool will be necessary for fuel line disconnection. Refer to Quick-Connect Fittings.
(6) Remove necessary vacuum lines at throttle body.
(7) Drain engine coolant and remove thermostat and thermostat housing.
(8) Remove PCV hose and valve at valve cover.
(9) Remove 3 upper intake manifold mounting bolts (Fig. 6), but only loosen 2 lower bolts about 2 turns.
(10) Disconnect 2 main engine harness connectors at rear of intake manifold (Fig. 5).

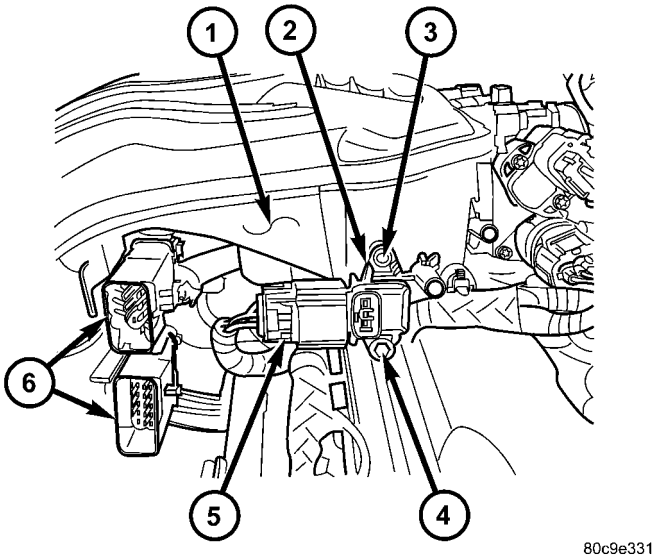

Fig. 5 MAP SENSOR LOCATION - 2.4L

1 - REAR OF INTAKE MANIFOLD
2 - MAP SENSOR
3 - ALIGNMENT PIN
4 - MOUNTING BOLT (TORX)
5 - ELECTRICAL CONNECTOR
6 - MAIN ENGINE HARNESS CONNECTORS